

Highlights

SAT-IT: Stability-aware Adversarial Training via Intermediate Trajectories

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- SAT-IT exploits intermediate PGD trajectories for adversarial training.
- A trajectory MI score measures sample-wise prediction instability.
- Stability-aware weighting emphasizes vulnerable samples adaptively.
- Lightweight self-distillation stabilizes robust optimization.
- Experiments show gains under PGD, CW_∞ , and AutoAttack.

SAT-IT: Stability-aware Adversarial Training via Intermediate Trajectories

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Abstract

Adversarial training has been demonstrated as an effective approach to enhance the robustness of deep neural networks against adversarial attacks. However, enhancing robustness inevitably leads to a decline in accuracy. We aim to enhance robustness while preserving accuracy, and believe the intermediate states in adversarial sample generation positively contribute to this goal—an overlooked aspect in existing research. To address this, we propose SAT-IT, a Stability-aware Adversarial Training framework via Intermediate Trajectories. SAT-IT introduces a mutual-information-based stability metric, measuring shifts in prediction distributions (intermediate trajectories). Using this metric, we develop an adaptive weighted robust training strategy that adaptively imposes stronger robust constraints on prediction-unstable samples while avoiding excessive interference on stable ones. Additionally, we incorporate a lightweight self-distillation mechanism to alleviate overfitting from stability-guided training and to guide the model towards smoother and more robust decision boundaries in later stages. Experimental results show that, compared to various existing adversarial training methods, SAT-IT improves model robustness while effectively mitigating the decline in natural accuracy.

Keywords: Adversarial training, Adversarial robustness, Intermediate trajectories, Mutual information, Sample reweighting, Self-distillation

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1. Introduction

In recent years, deep neural networks (DNNs) have achieved remarkable advancements in computer vision, speech recognition, and natural language processing. However, DNNs are highly vulnerable to adversarial attacks: attackers can induce erroneous model predictions by introducing imperceptible perturbations to the input data (Zhao et al., 2023).

Adversarial training (AT) is widely recognized as a highly effective method for enhancing robustness against adversarial attacks. It incorporates adversarial examples into training and models robustness as a min-max optimization problem, effectively boosting neural networks’ robustness (Madry et al., 2018; Dong et al., 2023; Qian et al., 2022). The most prevalent approach is projected gradient descent (PGD)-based AT (PGD-AT), which generates adversarial examples through multiple perturbation steps and updates parameters accordingly to bolster attack resistance. Building upon PGD-AT, a series of methods have been proposed (Wang et al., 2020; Wu et al., 2020; Jia et al., 2022). Nevertheless, these methods still face the robustness-accuracy trade-off: improving robustness against adversaries usually leads to performance degradation on clean samples (Zhang et al., 2019; Tsipras et al., 2019; Yang et al., 2020), limiting their practical application and broader adoption.

Most existing robust training methods primarily rely on the final adversarial examples generated by multi-step PGD, typically using the loss computed from the final states. In contrast, the intermediate samples produced along the PGD trajectory are largely overlooked, even though they capture the evolution of the predictive distribution.

However, different samples demonstrate notably divergent prediction dynamics along the PGD trajectory: certain samples traverse the decision boundary at intermediate steps, while others maintain correct classification yet experience substantial distribution shifts.

Motivated by these observations, we quantify sample-wise stability from the predictive sequence along the trajectory. Specifically, we use a mutual-information-based (MI-based) metric to capture the consistency (or fluctuation) of predictive distributions across $\{x_i^{(t)}\}$, and introduce a misclassification-event indicator to emphasize attack-prone samples. These two signals together enable an adaptive, sample-weighted robust training strategy.

Several recent works combine knowledge distillation with AT to improve the robustness-accuracy trade-off (Goldblum et al., 2020; Zi et al., 2021; Lu et al., 2025). In our experiments, dual-teacher distillation can produce

conflicting supervision: the natural teacher and adversarial teacher often disagree, leading to mismatched decision boundaries and unstable optimization. With a single teacher, the student tends to be teacher-limited in later stages, and the final performance is bounded by the teacher’s capacity. We therefore use a lightweight self-distillation scheme with label smoothing and an exponential moving average (EMA) of historical predictions to provide stable soft targets without external teachers (Li et al., 2024; Bai et al., 2024; Huang et al., 2023a).

We name our method SAT-IT, which stands for **S**tability-aware **A**dversarial **T**raining via **I**ntermediate **T**rajectories. The contributions of SAT-IT are twofold:

- **We propose a MI-based stability metric, integrating intermediate states into training for improved stability and robustness.**

We point out that existing AT methods ignore intermediate-state information during PGD adversarial example generation, and propose a MI-based, sample-level stability metric that leverages predictive distribution variations along the adversarial trajectory to guide robust training.

- **For stable training, we devise a stability-aware weighted AT strategy with lightweight self-distillation to ease overfitting.**

Based on the proposed stability metric, we design a stability-aware weighted AT strategy and integrate it with a lightweight self-distillation mechanism to alleviate potential overfitting induced by stability-driven training. Experimental results demonstrate the effectiveness of our method and its robustness advantages under various attack settings.

2. Related Work

Sample-wise Reweighting in Adversarial Training. Existing AT methods improve robustness via loss design, attack strategies, or structural optimization (Wang et al., 2020; Cui et al., 2024; Jia et al., 2022; Wu et al., 2020; Rice et al., 2020; Dong et al., 2023). To leverage sample heterogeneity, several works adopt sample-wise reweighting: MART incorporates sample difficulty into the robust objective (Wang et al., 2020); others address class imbalance in AT (Wang et al., 2022) or refine training dynamics via reweighting historical optimization trajectories (Huang et al., 2023b).

However, these methods rely on *static, single-point signals* (e.g., final adversarial loss, optimization history) and fail to capture the *dynamic predictive behavior* of samples during adversarial generation. They cannot distinguish samples that stay correctly classified but exhibit significant predictive distribution fluctuations. Complementing and extending these efforts, we exploit *intermediate trajectories* of adversarial generation to characterize per-sample instability, enabling finer-grained adaptive reweighting that targets truly vulnerable samples without over-constraining stable ones.

Friendly Adversarial Training (FAT) (Zhang et al., 2020) is closely related to our work because it also exploits intermediate adversarial states during PGD generation. Instead of always using the most adversarial examples, FAT searches for *friendly adversarial examples* by early stopping the PGD process once confidently misclassified samples are found. This strategy alleviates the negative impact of overly strong adversarial examples and helps preserve natural accuracy. However, FAT mainly uses intermediate states to select a single early-stopped adversarial endpoint for training. In contrast, SAT-IT does not treat intermediate states merely as candidate adversarial endpoints, but uses the entire prediction trajectory as a stability signal for adaptive sample-wise robust weighting.

Mutual Information in Machine Learning. Mutual information (MI) is a fundamental tool for quantifying predictive uncertainty and representation quality in machine learning (Gawlikowski et al., 2023). In adversarial robustness, MI has been used to construct robustness-oriented objectives (Zhou et al., 2022), balance natural and adversarial prediction uncertainty (Roh et al., 2020), or detect adversarial inputs via representation learning (Gao et al., 2023).

These works primarily apply MI to *model-level uncertainty* (e.g., parameter posterior) or *feature-level discrimination*, but overlook its potential to capture *sample-level dynamic instability* along adversarial trajectories. Inspired by MI’s effectiveness in uncertainty quantification, we propose a trajectory-aware MI metric that quantifies predictive distribution variations across PGD iterations. Unlike complex MI estimation in prior works, our metric is lightweight and directly computable from intermediate prediction sequences, serving as a real-time signal for adaptive sample weighting.

Adversarial Robust Distillation. Adversarially robust distillation (ARD) transfers robustness from a pre-trained robust teacher to a student model. Goldblum et al. showed that clean-data distillation can inherit teacher robustness

(Goldblum et al., 2020), while subsequent works improve teacher-student alignment via feature covariance matching (Dong et al., 2024) or multi-teacher fusion (combining robust and natural teachers) (Zhao et al., 2024, 2022; Lu et al., 2025). However, multi-teacher distillation suffers from supervision conflicts, and single-teacher methods limit student performance to the teacher’s capacity. Self-distillation has emerged as an alternative to stabilize training and alleviate robust overfitting via dynamic soft label refinement (Li et al., 2024).

Building on these regularization paradigms, we design a lightweight self-distillation mechanism that constructs stable soft targets from the student’s own historical predictions—eliminating external teacher dependence and cross-model conflicts. This mechanism complements our trajectory-based reweighting by suppressing over-sharpening of predictive distributions and enhancing training stability.

3. Methodology

We introduce a stability-aware AT framework that leverages intermediate samples along the PGD trajectory. Specifically, we *(i)* formalize adversarial sample trajectories, *(ii)* quantify sample-wise stability via a MI-based metric and use it to build adaptive robust weighting, and *(iii)* incorporate a lightweight self-distillation mechanism to improve generalization. An overview of the proposed SAT-IT framework is shown in Fig. 2.

3.1. Adversarial Sample Trajectories and Problem Formulation

This section introduces the adversarial example generation procedure used in this paper. We adopt a $K=10$ -step Projected Gradient Descent method to construct adversarial examples under an ℓ_∞ constraint. As shown in Eq. (1), during the PGD-10 generation process, in addition to the final adversarial example $x_i^{(K)}$, PGD also produces $K-1$ intermediate samples $\{x_i^{(1)}, \dots, x_i^{(K-1)}\}$. However, under the conventional min-max adversarial training framework, these intermediate samples are typically not used in the loss computation; training relies only on the initial sample $x_i^{(0)}$ and the final adversarial sample $x_i^{(K)}$.

$$x_i^{(t+1)} = \Pi_{\mathcal{B}_\varepsilon(x_i)} \left(x_i^{(t)} + \alpha \cdot \text{sign} \left(\nabla_x \mathcal{L}(f_\theta(x_i^{(t)}), y_i) \right) \right), \quad (1)$$

$$t = 0, \dots, K-1.$$

We observe that a model’s predictions for the same sample often exhibit complex and dynamic behavior during adversarial example generation. For a single sample x_i , the prediction sequence along the PGD iterations can be written as

$$\begin{aligned} p_i^{(t)} &= \text{softmax}(f_\theta(x_i^{(t)})), \\ y_i^{(t)} &= \text{argmax } p_i^{(t)}, \quad t = 0, \dots, K, \end{aligned} \tag{2}$$

where $p_i^{(t)}$ denotes the predicted probability vector (softmax output) on the intermediate sample $x_i^{(t)}$, and $y_i^{(t)}$ is the corresponding predicted label. Accordingly, we denote the probability trajectory as $\mathcal{P}_i = \{p_i^{(0)}, p_i^{(1)}, \dots, p_i^{(K)}\}$ and the label trajectory as $\mathcal{Y}_i = \{y_i^{(0)}, y_i^{(1)}, \dots, y_i^{(K)}\}$.

In empirical observations, model predictions along the PGD iterative trajectory exhibit substantial diversity. For one group of samples, the predicted class remains unchanged throughout the iterations and stays consistent with the ground-truth label. For another group, misclassification events occur at some point along the trajectory (i.e., the prediction becomes inconsistent with the true label), indicating that the decision boundary is crossed within the adversarial neighborhood. Moreover, even when the predicted class does not change, the trajectory can still exhibit pronounced fluctuations in the predicted probability distribution, reflecting varying degrees of neighborhood instability.

Many existing variants of AT characterize robustness using single-point signals (e.g., the terminal adversarial example or the prediction at a fixed iteration), and thus struggle to distinguish different cases such as “samples whose trajectories have already crossed the decision boundary” versus “samples whose class remains unchanged but whose predictive distribution fluctuates significantly.” This often leads to coarse-grained sample-wise robustness constraints. (Wang et al., 2020; Lu et al., 2025) To address this limitation, we introduce trajectory-based stability modeling: we use whether any misclassification occurs along the trajectory as an event-gating signal, and simultaneously employ the trajectory MI $I(Y; T \mid x_i)$ to continuously quantify the overall magnitude of predictive-distribution variation across the full trajectory. In this way, we achieve a finer-grained characterization of sample-level robustness and enable weighted training accordingly. As illustrated in Fig. 1, prediction dynamics along the trajectory can be highly diverse. We use top-1 prediction flips only as a qualitative visualization of such dynamics, whereas our stability modeling and weighting are based on misclassification

events and $I(Y; T \mid x_i)$.

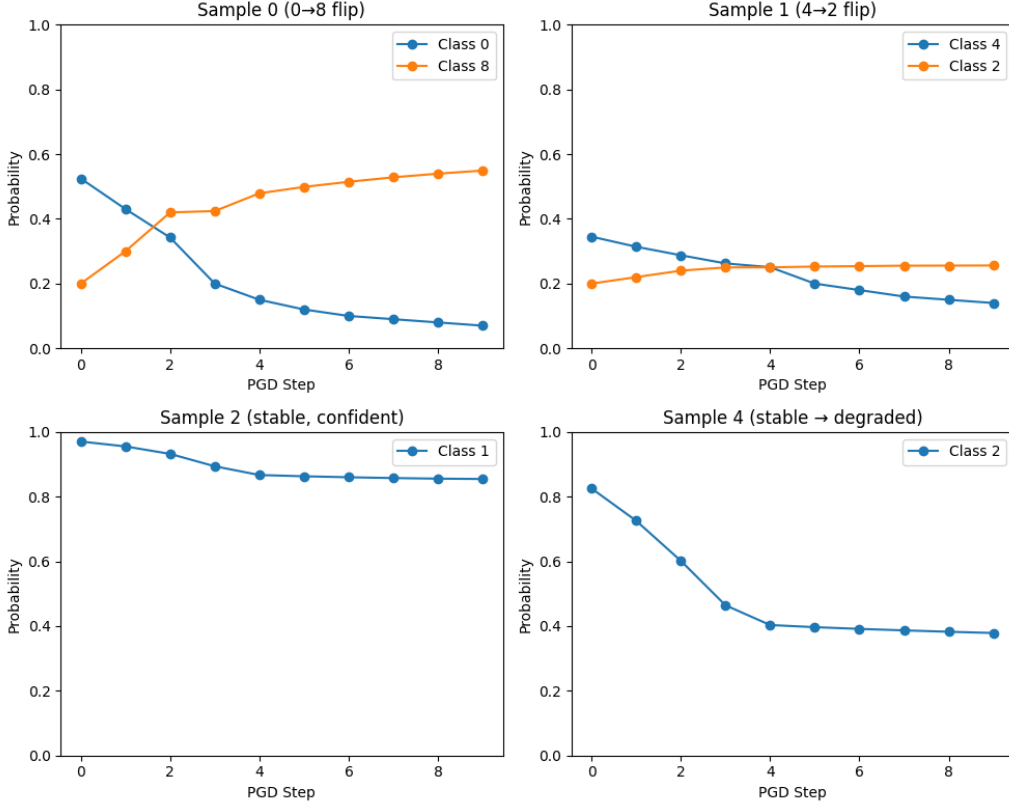


Figure 1: We present four typical PGD trajectory patterns: *early flip* (an early top-1 class switch), *mid flip* (a switch in the middle or late iterations), *stable-confident* (stable predictions with high confidence), and *stable→degraded* (stable class but gradually decreasing confidence). Here, “flip” denotes a top-1 prediction class switch along PGD iterations (e.g., 0→8), used only for qualitative visualization.

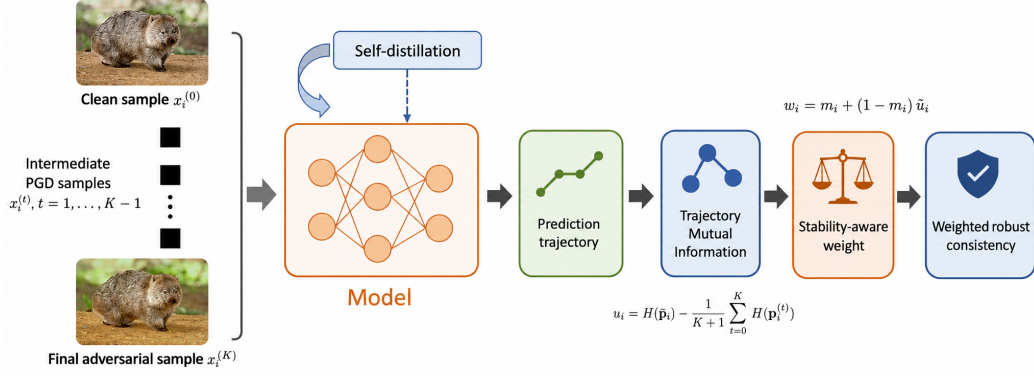


Figure 2: Overview of the proposed SAT-IT framework. We generate intermediate samples along the PGD trajectory, quantify trajectory stability via MI indicator, and apply stability-aware weighting in the outer robust objective (with entropy regularization)

3.2. Stability Characterization via Mutual Information

Stability and uncertainty estimation are closely related in deep models, and prior surveys have systematically summarized relevant methods (Gawlikowski et al., 2023; Belghazi et al., 2018). MI, a core metric in information theory, characterizes statistical dependence between random variables. In deep models, it is commonly used to quantify predictive uncertainty under a specified randomness source, and can also serve as a principled measure of predictive disagreement/variation across multiple predictive distributions (Roh et al., 2020; Zhou et al., 2022).

In classification tasks, MI is often used to characterize predictive uncertainty (or variability) by measuring the dependence between the predicted label Y and a source of randomness Z (e.g., model parameters in Bayesian settings) (Houlsby et al., 2011). In BALD (Bayesian Active Learning by Disagreement), $I(Y; Z)$ quantifies predictive disagreement induced by parameter uncertainty, i.e., the reduction in label uncertainty when conditioning on a particular realization of Z . Its core form can be expressed as the predictive entropy minus the expected conditional entropy (Cao and Tsang, 2021). Equation (3) represents the general form used by this method.

$$I(Y; Z) = H(Y) - \mathbb{E}_{z \sim p(z)}[H(Y | Z = z)]. \quad (3)$$

We adopt the same MI decomposition, but since AT cares more about prediction stability within the local neighborhood of an input, we replace

the “source of randomness” in MI from the parameter posterior θ to the trajectory state variable T along adversarial example generation. This leads to a sample-wise stability metric $I(Y; T \mid x_i)$. Here we do not interpret $I(Y; T \mid x_i)$ as epistemic uncertainty; rather, it measures trajectory-induced predictive variation/disagreement across PGD states.

We define T as a discrete latent variable indicating the PGD iteration step along the trajectory ($t \in \{0, \dots, K\}$), where K is the maximum number of perturbation steps. We treat the trajectory index as a discrete latent variable and measure how predictive distributions vary when uniformly sampling states along the PGD path; accordingly, we assume a uniform prior

$$p(t) = \frac{1}{K+1}. \quad (4)$$

Given an input x_i , the trajectory samples $x_i^{(t)}$ are generated by PGD. At each state, the model outputs a predictive distribution $p_\theta(y \mid x_i^{(t)})$, which induces the joint distribution

$$p(y, t \mid x_i) = p(t) p_\theta(y \mid x_i^{(t)}). \quad (5)$$

Consequently, MI can be written as

$$I(Y; T \mid x_i) = H(\bar{p}_i) - \mathbb{E}_t \left[H(p_i^{(t)}) \right], \quad (6)$$

where $\bar{p}_i = \mathbb{E}_t[p_i^{(t)}]$ and $p_i^{(t)} = p_\theta(y \mid x_i^{(t)})$. This quantity measures the degree of dispersion of the predictive distributions as the trajectory state varies. For clarity, we refer to it as a *trajectory instability* score throughout the paper (smaller is more stable).

$$\begin{aligned} I(Y; T \mid x_i) &= H(\bar{p}_i) - \frac{1}{K+1} \sum_{t=0}^K H(p_i^{(t)}), \\ \bar{p}_i &= \frac{1}{K+1} \sum_{t=0}^K p_i^{(t)}. \end{aligned} \quad (7)$$

where $p_i^{(t)} = p_\theta(y \mid x_i^{(t)})$ denotes the predictive distribution at step t .

Based on $\{p_i^{(t)}\}_{t=0}^K$, we compute the entropy terms in the MI decomposition.

The corresponding entropy of the trajectory-averaged predictive distribution is

$$H(\bar{\mathbf{p}}_i) = - \sum_{c=1}^C \bar{p}_{i,c} \log \bar{p}_{i,c}, \quad (8)$$

where $\bar{\mathbf{p}}_i \in \mathbb{R}^C$ denotes the average predictive probability vector along the trajectory, and $\bar{p}_{i,c}$ is its c -th component.

The expected conditional entropy is

$$\mathbb{E}_t \left[H(\mathbf{p}_i^{(t)}) \right] = \frac{1}{K+1} \sum_{t=0}^K \left(- \sum_{c=1}^C p_{i,c}^{(t)} \log p_{i,c}^{(t)} \right), \quad (9)$$

where $\mathbf{p}_i^{(t)} = \text{softmax}(f_\theta(x_i^{(t)}))$ is the predictive probability vector at step t .

Therefore, the MI-based indicator for sample x_i is defined as

$$u_i = H(\bar{\mathbf{p}}_i) - \frac{1}{K+1} \sum_{t=0}^K H(\mathbf{p}_i^{(t)}). \quad (10)$$

This indicator characterizes the prediction instability of an individual sample within the adversarial neighborhood centered at x_i with radius ε .

Intuitively, $H(\bar{\mathbf{p}}_i)$ reflects the overall *trajectory-induced variation/inconsistency* of the aggregated prediction, while $\frac{1}{K+1} \sum_{t=0}^K H(\mathbf{p}_i^{(t)})$ measures the average per-state predictive *entropy* along the trajectory.

For stable samples, $H(\bar{\mathbf{p}}_i)$ is close to $\frac{1}{K+1} \sum_{t=0}^K H(\mathbf{p}_i^{(t)})$, resulting in a small u_i , and we regard them as stable. In contrast, samples with larger u_i can be viewed as more vulnerable (attack-prone) samples.

3.3. Stability-Aware Sample Weighting

Based on the MI-based *instability* indicator u_i defined in the previous section, we incorporate it into the AT objective and propose a stability-aware sample reweighting strategy (SAT-IT) to guide the allocation of sample-wise robustness constraints. The core idea is to impose stronger robustness regularization on prediction-unstable samples (with larger u_i), while avoiding excessive intervention on stable samples, thereby alleviating the robustness–accuracy trade-off in AT.

Along the PGD-generated neighborhood, samples exhibit heterogeneous robustness behaviors. Some samples become misclassified at least once within

the K PGD states, signaling a decision-boundary crossing. However, this binary event alone is coarse: even when the top-1 label stays correct for all states, the predictive distribution can still vary substantially (e.g., confidence erosion or mass shifting among competing classes), indicating different degrees of neighborhood-level instability.

Motivated by these observations, we use the MI-based instability indicator u_i in Eq. (10) as a continuous metric of trajectory-level predictive-distribution variation, and combine it with a binary misclassification gate m_i to construct sample-wise weights.

To make the scale comparable, we normalize u_i by min-max scaling within each mini-batch:

$$\tilde{u}_i = \frac{u_i - u_{\min}}{u_{\max} - u_{\min} + \tau}, \quad (11)$$

where $u_{\min}$ and $u_{\max}$ denote the minimum and maximum of $\{u_i\}$ in the current mini-batch, and τ is a small constant for numerical stability. We then define the final sample weight as

$$w_i = m_i + (1 - m_i) \cdot \tilde{u}_i, \quad (12)$$

where $m_i \in \{0, 1\}$ indicates whether x_i becomes misclassified at any step along the PGD trajectory.

As shown in Eq. (12), we decompose the sample-wise weight into two components: an *event gate* and a *continuous measure*. When a misclassification occurs along the trajectory ($m_i = 1$), it indicates that the sample has crossed the decision boundary within the adversarial neighborhood; thus we directly assign the maximum weight to strengthen the robustness constraint. When the sample remains correctly classified throughout the trajectory ($m_i = 0$), we instead use the normalized instability score $\tilde{u}_i$ to adaptively quantify the magnitude of prediction-distribution fluctuations, thereby distinguishing *stable and high-confidence* samples from *always-correct* samples whose distributions still vary substantially.

$$w_i = \begin{cases} 1, & m_i = 1, \\ \tilde{u}_i \in [0, 1], & m_i = 0. \end{cases} \quad (13)$$

In implementation, the trajectory-based weights are computed online and detached from back-propagation. They are used only as sample-wise coefficients in the outer objective, avoiding additional gradient paths through the PGD trajectory statistics.

At the objective level, the proposed stability-aware reweighting strategy is applied only to the *outer* minimization in AT and does not change the underlying min-max structure. Specifically, adversarial examples are still generated by approximately solving the inner maximization with standard PGD:

$$x_i^{\text{adv}} = \arg \max_{\|\delta\| \leq \varepsilon} \mathcal{L}(f_\theta(x_i + \delta), y_i), \quad (14)$$

where the inner optimization process is identical to conventional AT.

In the outer minimization stage, we introduce the stability weight w_i into the robust consistency term to impose different strengths of robustness constraints across samples. Specifically, we define the weighted robust consistency loss as

$$\mathcal{L}_{\text{rob}} = \frac{1}{B} \sum_{i=1}^B w_i \text{KL}(p_i^{\text{adv}} \| p_i^{\text{nat}}), \quad (15)$$

where B is the mini-batch size. The overall training objective is formulated as

$$\mathcal{L} = \mathcal{L}_{\text{sd}} + \mathcal{L}_{\text{adv}} + \beta \mathcal{L}_{\text{rob}} - \lambda \mathcal{R}_{\text{ent}}, \quad (16)$$

where $\mathcal{L}_{\text{sd}}$ denotes the self-distilled clean loss computed on natural samples using historical soft targets, $\mathcal{L}_{\text{adv}}$ denotes the adversarial loss, and $\mathcal{L}_{\text{rob}}$ enforces robust consistency between predictions on adversarial and clean samples. The entropy-based regularizer $\mathcal{R}_{\text{ent}}$ is introduced to mitigate the over-sharpening of predictive distributions that may be induced by stability-aware weighting.

We define $\mathcal{R}_{\text{ent}}$ in a stage-wise manner as

$$\mathcal{R}_{\text{ent}} = \begin{cases} \frac{1}{B} \sum_{i=1}^B w_i H(p_\theta(\cdot | x_i)), & e < e_0, \\ \frac{1}{B} \sum_{i=1}^B w_i H(p_\theta(\cdot | x_i^{\text{adv}})), & e \geq e_0, \end{cases} \quad (17)$$

where e denotes the training epoch and e_0 is the stage switching point. In the early stage, $\mathcal{R}_{\text{ent}}$ is applied to predictions on clean samples to suppress over-confident outputs. As training proceeds and the decision boundary gradually stabilizes, $\mathcal{R}_{\text{ent}}$ is instead applied to predictions on adversarial samples to counteract the overly peaked distributions caused by the combined effects of stability-aware weighting and robust consistency constraints. This stage-wise design helps balance discriminability and smoothness without weakening robustness.

Algorithm 1 SAT-IT

Input: mini-batch $\{(x_i, y_i)\}_{i=1}^B$; model f_θ ; PGD (K, ε, α) ; hyper-params $(\beta, \lambda, a, b, c)$; switch epoch e_0 .

Output: total loss $\mathcal{L}$.

- 1: Generate PGD trajectory $\{x_i^{(t)}\}_{t=0}^K$ by Eq. (1).
 - 2: Collect $p_i^{(t)} = \text{softmax}(f_\theta(x_i^{(t)}))$ and $y_i^{(t)} = \arg \max p_i^{(t)}$ (Eq. (2)).
 - 3: Compute misclassification gate $m_i = \mathbb{I}[\exists t : y_i^{(t)} \neq y_i]$.
 - 4: Compute $\bar{\mathbf{p}}_i = \frac{1}{K+1} \sum_{t=0}^K \mathbf{p}_i^{(t)}$ and MI instability u_i by Eq. (10).
 - 5: Normalize $\tilde{u}_i$ by Eq. (11) and compute detached weights w_i by Eq. (12).
 - 6: Compute adversarial loss $\mathcal{L}_{\text{adv}}$ on terminal adversarial samples $\{(x_i^{(K)}, y_i)\}$.
 - 7: Compute weighted robust consistency $\mathcal{L}_{\text{rob}}$ by Eq. (15).
 - 8: Compute stage-wise entropy regularizer $\mathcal{R}_{\text{ent}}$ by Eq. (17) (switch at epoch e_0).
 - 9: Form mixed prediction $\mathbf{p}_i^{\text{mix}}$ (Eq. (18)), update the self-distillation target $\mathbf{q}_i$ using historical predictions (Eq. (19)), and construct soft target $\mathbf{y}_i^{\text{soft}}$ (Eq. (20)).
 - 10: Compute self-distilled clean loss $\mathcal{L}_{\text{sd}}$ by Eq. (21).
 - 11: **return** $\mathcal{L} = \mathcal{L}_{\text{sd}} + \mathcal{L}_{\text{adv}} + \beta \mathcal{L}_{\text{rob}} - \lambda \mathcal{R}_{\text{ent}}$.
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3.4. Self-Distillation as a Collaborative Constraint

In recent years, several studies have incorporated knowledge distillation into AT to mitigate the robustness–accuracy trade-off. However, teacher–student distillation typically requires balancing objectives among a natural teacher, a robust teacher, and the student model.

$$\mathbf{p}_i^{\text{mix}} = c \mathbf{p}_i^{\text{nat}} + (1 - c) \mathbf{p}_i^{\text{adv}}, \quad (18)$$

$$\mathbf{q}_i \leftarrow a \mathbf{q}_i + (1 - a) \text{stopgrad}(\mathbf{p}_i^{\text{mix}}), \quad (19)$$

$$\mathbf{y}_i^{\text{soft}} = b \mathbf{q}_i + (1 - b) \mathbf{y}_i^{\text{onehot}}, \quad (20)$$

$$\mathcal{L}_{\text{sd}} = -\frac{1}{B} \sum_{i=1}^B \sum_{k=1}^C y_{i,k}^{\text{soft}} \log(\text{softmax}(\mathbf{z}_i^{\text{nat}})_k), \quad (21)$$

To avoid introducing external teacher models and eliminate cross-model supervision conflicts, we adopt a lightweight *self-distillation* mechanism that

constructs soft supervision purely from the student’s own historical predictions. Specifically, we first form a mixed prediction distribution by combining natural and adversarial predictive probabilities (Eq. (18)), and then update a historical self-distillation target via an exponential moving average of the mixed probabilities with a stop-gradient operator (Eq. (19)). Next, we blend this historical target with the one-hot ground-truth label to obtain the final soft target $\mathbf{y}^{\text{soft}}$ (Eq. (20)). Finally, we compute the self-distilled clean loss $\mathcal{L}_{\text{sd}}$ on natural logits using $\mathbf{y}^{\text{soft}}$ (Eq. (21)).

This self-distillation provides stable, history-smoothed supervision without cross-model mismatch. In the final objective, it replaces the ordinary one-hot clean cross-entropy term rather than being added as an extra independent loss term. Therefore, SAT-IT suppresses output over-sharpening and reduces training instability while keeping the standard inner-loop PGD procedure unchanged.

4. Experiments

4.1. Experimental Setup

Datasets & Models. To validate the effectiveness of our method, we conduct experiments on CIFAR-10, CIFAR-100, and TinyImageNet-200. CIFAR-10 and CIFAR-100 are used as standard image classification benchmarks, while TinyImageNet-200 is further adopted to evaluate the scalability of SAT-IT on a more challenging dataset with more classes and higher-resolution images. For robust training on CIFAR datasets, we adopt MobileNetV2 as a lightweight student model and ResNet-18 as a commonly used convolutional baseline, covering architectures with different capacities. For TinyImageNet-200, we use ResNet-18 for evaluation. For distillation-based settings, we train the teacher using WideResNet-34-10, and construct both a natural teacher and a robust teacher to provide supervision signals to the student. Teachers are trained only for distillation baselines; SAT-IT itself does not use any external teacher.

Parameter Setting. Unless otherwise specified, all methods are trained under the same setup. We train for 120 epochs under an ℓ_∞ budget $\varepsilon = 8/255$, and generate adversarial examples using PGD-10 (step size $2/255$). We use a mini-batch size of 128 for training and 100 for validation, and fix the PyTorch random seed to 0. We use SGD with momentum 0.9 and a stepwise learning-rate schedule decayed by $10\times$ at epochs 75, 90, and 100; the initial learning

rate / weight decay are $0.01 / 3.5 \times 10^{-3}$ for non-distillation methods and $0.1 / 2 \times 10^{-4}$ for distillation-based methods. All experiments are implemented in PyTorch and run on a single NVIDIA RTX 4090 GPU. For MTARD and B-MTARD, we scale the robust loss by $\beta=10$ for training stability.

Evaluation Metrics. We report the standard (clean) accuracy and robust accuracy under multiple white-box attacks, including FGSM, PGD-20, PGD-100, CW_∞ , and AutoAttack. For CW_∞ , we use the standard setup with confidence=50, lr=2/255, steps=30, and set `num_classes`= C for each dataset. For PGD-20 and PGD-100, we evaluate under the ℓ_∞ threat model with $\varepsilon = 0.031$ and `step_size`= 0.003, where `num_steps` is set to 20 and 100, respectively. For AutoAttack, we use the official implementation, which includes APGD-CE, APGD-DLR, FAB, and Square Attack, and report the aggregated result as the final robustness metric.

Table 1: Results on CIFAR-10 with MobileNetV2 (best epoch).

Method	Clean	FGSM	PGD-20	PGD-100	CW_∞	AutoAttack
AT	0.8324	0.5710	0.5252	0.4953	0.4953	0.4649
TRADES	0.8238	0.5729	0.5338	0.5140	0.4886	0.4768
MART	0.7782	0.5704	0.5482	0.5348	0.4804	0.4645
RSLAD	0.8090	0.5619	0.5239	0.5014	0.4838	0.4643
MTARD	0.8538	0.5713	0.5074	0.4734	0.4567	0.4293
B-MTARD	0.8415	0.5544	0.5040	0.4817	0.4632	0.4448
FAT	0.8416	0.5568	0.5040	0.4889	0.4507	0.4320
SAT-IT (Ours)	0.7879	0.5740	0.5504	0.5362	0.4843	0.4705

Table 2: Results on CIFAR-100 with MobileNetV2 (best epoch).

Method	Clean	FGSM	PGD-20	PGD-100	CW_∞	AutoAttack
AT	0.5641	0.3211	0.3002	0.2845	0.2659	0.2381
TRADES	0.5971	0.3213	0.2990	0.2873	0.2510	0.2334
MART	0.4945	0.3218	0.3114	0.3041	0.2592	0.2416
RSLAD	0.4839	0.3204	0.3099	0.3029	0.2708	0.2525
MTARD	0.5811	0.3142	0.2840	0.2704	0.2406	0.2177
B-MTARD	0.5074	0.3182	0.3041	0.2960	0.2543	0.2373
FAT	0.6010	0.2854	0.2523	0.2340	0.2188	0.1954
SAT-IT (Ours)	0.5259	0.3377	0.3255	0.3176	0.2679	0.2490

4.2. Effectiveness of SAT-IT

We evaluate SAT-IT under the standard white-box threat model.

As shown in Tables 1 and 2, we compare SAT-IT with AT (Madry et al., 2018), TRADES (Zhang et al., 2019), MART (Wang et al., 2020), RSLAD (Zi et al., 2021), MTARD (Zhao et al., 2022), and B-MTARD (Zhao et al., 2024). On the lightweight MobileNet-V2 model, SAT-IT achieves robustness gains under white-box attacks and exhibits a favorable robustness–accuracy trade-off. Specifically, at the best epoch (97), SAT-IT attains robust accuracies of 0.5504/0.5362 under PGD-20/PGD-100 and 0.4705 under AutoAttack, being competitive with or better than AT baselines. In particular, compared with MART, SAT-IT consistently yields higher performance across all evaluated robustness metrics, suggesting that stability-aware sample reweighting based on trajectory stability can further strengthen robustness under strong attacks without introducing additional model-architecture modifications.

On the more challenging CIFAR-100 benchmark, the improvements of SAT-IT are more pronounced. SAT-IT achieves 0.3377/0.3255/0.3176 under FGSM/PGD-20/PGD-100 and 0.2490 under AutoAttack. Compared with TRADES, SAT-IT exhibits clear advantages on strong-attack metrics, e.g., AutoAttack 0.2490 vs. 0.2334, and PGD-100 0.3176 vs. 0.2873. These results indicate that when the dataset becomes more fine-grained and complex, trajectory-based stability modeling with sample-wise reweighting can better concentrate robustness constraints on truly vulnerable samples, whereas TRADES tends to be less effective under this setting. A similar trend is also observed for ResNet-18 on CIFAR-100.

Compared with the robust distillation baseline RSLAD, SAT-IT performs better on CIFAR-10 under strong iterative attacks (e.g., higher PGD-20/100 and AutoAttack), and on CIFAR-100 it achieves higher clean accuracy while also outperforming RSLAD on PGD-20/100. Overall, these results demonstrate that trajectory stability modeling and sample-wise reweighting can effectively improve the robustness of lightweight models under strong iterative attacks *without* relying on teacher models.

Furthermore, we compare SAT-IT with FAT, which also exploits intermediate adversarial states but uses them mainly to construct early-stopped friendly adversarial examples. As shown in Table 4, FAT achieves higher clean accuracy on CIFAR-100 with ResNet-18, but its robustness under strong attacks is substantially lower than SAT-IT. Specifically, SAT-IT improves over FAT from 0.2819 to 0.3381 under PGD-20, from 0.2577 to 0.3288 under PGD-100, and from 0.2261 to 0.2608 under AutoAttack. These results suggest that simply selecting a friendly intermediate adversarial endpoint is insufficient for strong robustness, while modeling the full adversarial trajec-

tory and using it for stability-aware weighting provides more effective robust supervision.

Table 3: Results on CIFAR-10 with ResNet-18 (best epoch).

Method	Clean	FGSM	PGD-20	PGD-100	CW _∞	AutoAttack
Natural	0.9497	0.2962	0	0	0	0
AT	0.8400	0.5805	0.5378	0.5137	0.5078	0.4850
TRADES	0.8325	0.5829	0.5438	0.5208	0.5015	0.4881
MART	0.8085	0.5948	0.5622	0.5438	0.5023	0.4855
LBGAT	0.8279	0.5848	0.5555	0.5382	0.5119	0.4989
RSLAD	0.8155	0.5812	0.5385	0.5152	0.4982	0.4792
MTARD	0.8726	0.5811	0.5050	0.4671	0.4548	0.4305
B-MTARD	0.8568	0.5878	0.5201	0.4886	0.4703	0.4494
FAT	0.8641	0.5953	0.5194	0.4769	0.4862	0.4615
SAT-IT (Ours)	0.8291	0.6061	0.5727	0.5527	0.5028	0.4874

Table 4: Results on CIFAR-100 with ResNet-18 (best epoch).

Method	Clean	FGSM	PGD-20	PGD-100	CW _∞	AutoAttack
AT	0.5823	0.3294	0.3095	0.2946	0.2763	0.2523
TRADES	0.6088	0.3249	0.3020	0.2898	0.2522	0.2398
MART	0.5436	0.3427	0.3295	0.3207	0.2819	0.2631
MART+LORE	0.5474	0.3399	0.3263	0.3171	0.2783	0.2588
RSLAD	0.5084	0.3302	0.3189	0.3106	0.2750	0.2564
MTARD	0.6326	0.3123	0.2718	0.2509	0.2322	0.2055
B-MTARD	0.5396	0.3276	0.3057	0.2964	0.2595	0.2400
FAT	0.6536	0.3224	0.2819	0.2577	0.2482	0.2261
SAT-IT (Ours)	0.5650	0.3537	0.3381	0.3288	0.2797	0.2608

As shown in Tables 3–4, we compare our method with AT, TRADES, MART, LBGAT (Cui et al., 2021), RSLAD, MTARD, B-MTARD, and FAT. For CIFAR-100, we additionally include MART+LORE (Yin and Ruan, 2024) as a recent strong baseline. TRADES remains a strong baseline on CIFAR-10, but its robustness drops notably on CIFAR-100: its PGD-100/AutoAttack accuracies are only 0.2898/0.2398, whereas SAT-IT achieves 0.3288/0.2608. This larger cross-dataset gap suggests that TRADES is less stable when moving to finer-grained and more complex data distributions.

On CIFAR-10, SAT-IT consistently improves over MART across the reported robustness metrics: it attains 0.5727/0.5527 under PGD-20/PGD-100, higher than MART’s 0.5622/0.5438, and also yields a slightly better AutoAttack accuracy (0.4874 vs. 0.4855). These gains mainly manifest un-

der strong iterative attacks, indicating the effectiveness of trajectory stability modeling with sample-wise reweighting.

Compared with FAT, SAT-IT shows a clear robustness advantage under strong attacks. On CIFAR-10, FAT achieves higher clean accuracy (0.8641 vs. 0.8291), but SAT-IT improves the robust accuracy from 0.5194 to 0.5727 under PGD-20, from 0.4769 to 0.5527 under PGD-100, and from 0.4615 to 0.4874 under AutoAttack. A similar trend is observed on CIFAR-100: although FAT obtains higher clean accuracy (0.6536 vs. 0.5650), SAT-IT substantially improves PGD-20/PGD-100/AutoAttack robustness from 0.2819/0.2577/0.2261 to 0.3381/0.3288/0.2608. These results suggest that FAT’s early-stopped friendly adversarial examples help preserve clean accuracy, while SAT-IT provides stronger robustness by modeling the full adversarial trajectory and applying stability-aware weighting.

We also observe a clear trend for distillation-based methods. When switching from the small MobileNet-V2 to the medium-scale ResNet-18, the relative advantage of distillation becomes less pronounced. For example, the single-teacher distillation method RSLAD no longer leads on AutoAttack: it achieves 0.4792 on CIFAR-10 (below SAT-IT’s 0.4874) and 0.2564 on CIFAR-100 (below MART’s 0.2631). This suggests that distillation-based approaches are more sensitive to teacher quality and loss balancing, whereas SAT-IT does not rely on a teacher and exhibits more stable performance when changing architectures and datasets.

Figure 3 shows the training curves of SAT-IT and non-distillation AT baselines on CIFAR-10. SAT-IT maintains competitive robust accuracy during training and achieves a favorable robustness–accuracy trade-off.

Figure 4 further compares SAT-IT with teacher-distillation-based methods. The results show that SAT-IT remains stable without relying on external teacher models.

Transferability and scalability. To further evaluate the generalization of SAT-IT, we conduct black-box transfer attacks on CIFAR-10 and extend the evaluation to TinyImageNet-200. As shown in Table 5, SAT-IT achieves the best performance under PGD attacks transferred from TRADES and CW_∞ attacks transferred from TRADES, while remaining competitive under attacks transferred from MART. These results suggest that the robustness gains of SAT-IT are not limited to white-box evaluation.

We further evaluate SAT-IT on TinyImageNet-200, a more challenging dataset with higher resolution and more classes. As shown in Table 6, SAT-

Table 5: Black-box transfer attack results on CIFAR-10 with ResNet-18 (best epoch).

Method	Clean	PGD _{TRADES}	PGD _{MART}	CW _{∞,TRADES}	CW _{∞,MART}	Epoch
AT	0.8400	0.6365	0.6508	0.6234	0.6211	79
TRADES	0.8325	0.5395	0.6526	0.5136	0.6291	106
MART	0.8085	0.6254	0.5604	0.6171	0.5174	90
MART+LORE	0.8276	0.6341	0.6361	0.6227	0.6015	111
LBGAT	0.8279	0.6245	0.6509	0.6106	0.6234	112
RSLAD	0.8155	0.6219	0.6359	0.6142	0.6071	108
SAT-IT (Ours)	0.8284	0.6373	0.6462	0.6261	0.6109	94

Table 6: Results on TinyImageNet-200 with ResNet-18 (best epoch).

Method	Clean	PGD-10	FGSM	PGD-100	CW _∞	AutoAttack	Epoch
Natural	0.4545	0	–	–	–	–	–
TRADES	0.3785	0.1375	0.1460	0.1310	0.0905	0.0795	75
MART	0.3175	0.1690	0.1725	0.1650	0.1210	0.1085	79
LBGAT	0.1540	0.0755	$\approx 0^\dagger$	$\approx 0^\dagger$	$\approx 0^\dagger$	$\approx 0^\dagger$	96
RSLAD	0.2630	0.1605	0.1645	0.1560	0.1125	0.1030	112
MTARD	0.2295	0.1200	0.1305	0.1170	0.0880	0.0740	109
B-MTARD	0.2245	0.1335	0.1390	0.1310	0.0930	0.0830	84
FAT	0.3930	0.1205	0.1380	0.1075	0.0840	0.0730	98
SAT-IT (Ours)	0.3230	0.1700	0.1725	0.1660	0.1260	0.1160	77

[†] For LBGAT on TinyImageNet-200, the corresponding robust accuracies collapse to zero or near-zero under this setting; we mark them as ≈ 0 to indicate method failure/inapplicability on this dataset rather than unavailable favorable results.

IT consistently achieves the best robustness under PGD-10, PGD-100, CW_∞, and AutoAttack among robust training methods. In contrast, LBGAT shows a clear collapse on TinyImageNet-200: after PGD-10, the remaining robust accuracies are zero or near-zero in this setting. This suggests that LBGAT is not well suited to this more complex dataset configuration, whereas trajectory stability modeling remains effective when scaling to more complex visual recognition tasks.

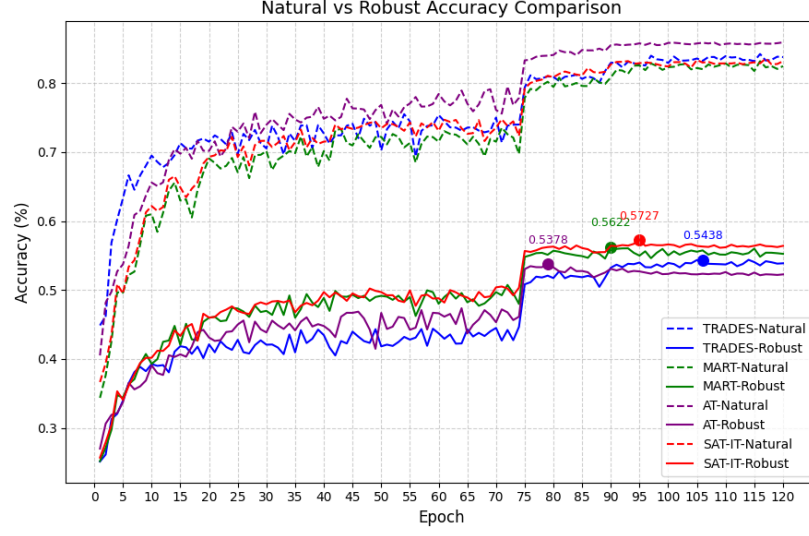


Figure 3: Training curves of ResNet-18 on CIFAR-10 over 120 epochs, comparing SAT-IT with *non-distillation* AT baselines. Dashed lines denote natural (clean) accuracy and solid lines denote robust accuracy. Dots mark the epoch achieving the best robust accuracy for each method.

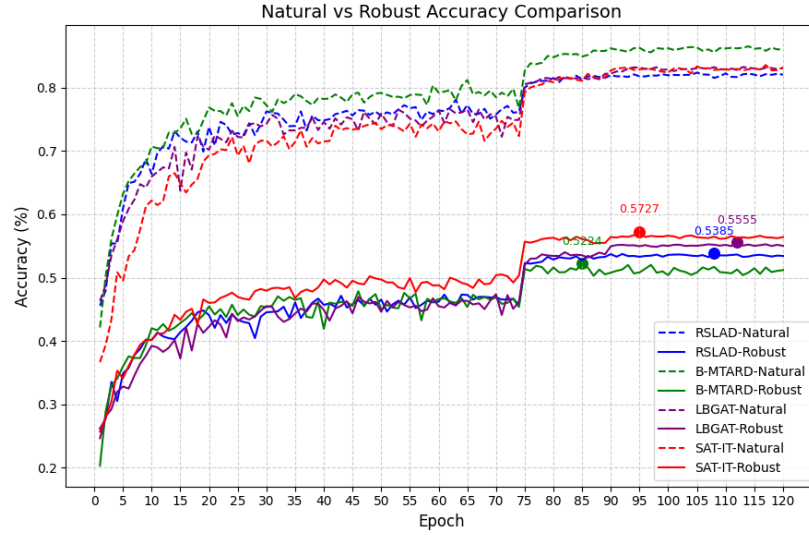


Figure 4: Training curves of ResNet-18 on CIFAR-10 over 120 epochs, comparing SAT-IT with *teacher-distillation* based methods. Dashed lines denote natural (clean) accuracy and solid lines denote robust accuracy. Dots mark the epoch achieving the best robust accuracy for each method.

4.3. Post-hoc Trajectory Stability Analysis

To further distinguish SAT-IT from FAT, we conduct a post-hoc trajectory stability analysis using trained ResNet checkpoints on CIFAR-10 and CIFAR-100. This analysis does not introduce additional training or modify the optimization procedure; instead, it evaluates the prediction dynamics of already trained models along the same PGD trajectory. Although both FAT and SAT-IT involve intermediate adversarial states, they exploit them in fundamentally different ways. FAT mainly uses intermediate states to select early-stopped friendly adversarial examples, aiming to avoid overly aggressive adversarial samples during training. In contrast, SAT-IT does not merely choose one intermediate endpoint. It explicitly models the full PGD prediction trajectory and uses the trajectory-level instability signal to guide sample-wise robust constraints.

Specifically, for each test sample, we generate a PGD trajectory and compute the trajectory MI score $I(Y; T \mid x_i)$ following Eq. (10). A smaller trajectory MI indicates that the predictive distribution changes less across PGD states, corresponding to more stable local prediction behavior. Conversely, a larger trajectory MI means that the model’s output distribution varies significantly as the adversarial perturbation evolves, suggesting that the sample lies in a more unstable neighborhood. Therefore, the trajectory MI distribution provides a direct way to examine whether a trained model has learned stable prediction behavior along adversarial paths.

As shown in Fig. 5, SAT-IT produces trajectory MI distributions more concentrated near zero on both CIFAR-10 and CIFAR-100, while FAT exhibits a more pronounced high-MI tail. This indicates that, for a larger portion of test samples, SAT-IT maintains more consistent predictive distributions throughout the PGD process. In contrast, FAT still leaves many samples with relatively large trajectory-induced prediction fluctuations, even though it uses intermediate adversarial examples during training. This difference supports our claim that intermediate states are not only useful as early-stopped adversarial endpoints, but can also serve as informative signals for modeling prediction stability.

These observations are consistent with the quantitative robustness results. FAT tends to preserve higher clean accuracy, which aligns with its design of using friendly adversarial examples. However, SAT-IT achieves stronger robustness under iterative and stronger attacks, especially PGD-20, PGD-100, and AutoAttack. The post-hoc MI analysis provides an explanation for this behavior: by explicitly reducing trajectory-level prediction instability,

SAT-IT learns a more stable local prediction structure around adversarial paths. Therefore, SAT-IT stabilizes the full adversarial trajectory rather than merely selecting an intermediate adversarial endpoint.

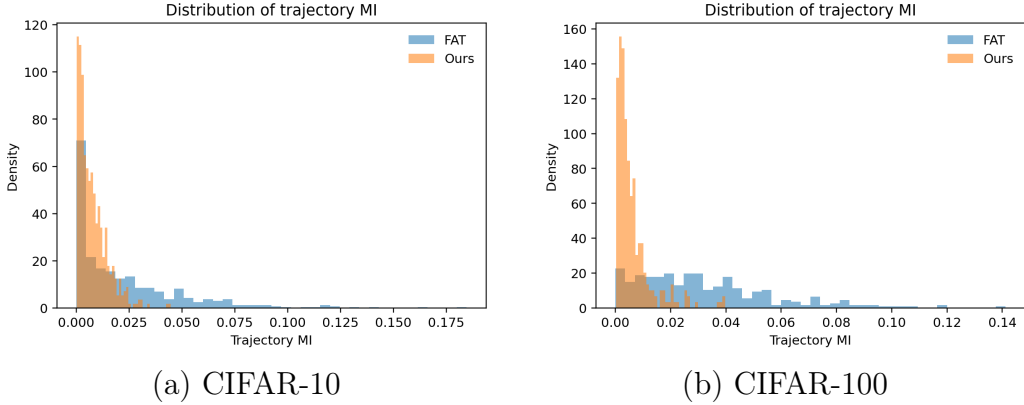


Figure 5: Post-hoc trajectory stability analysis on CIFAR-10 and CIFAR-100. We compare the distribution of trajectory MI between FAT and SAT-IT using trained checkpoints. Lower trajectory MI indicates more stable prediction distributions along PGD trajectories. Compared with FAT, SAT-IT concentrates more samples near zero and reduces high-instability samples.

We further visualize the penultimate-layer feature distributions under natural inputs, final PGD-20 adversarial inputs, and intermediate PGD trajectory states. As shown in Fig. 6, PGD trajectories induce noticeable feature shifts compared with natural inputs. Compared with FAT, SAT-IT exhibits more stable trajectory-induced feature evolution, which is consistent with the trajectory MI analysis.



Figure 6: PCA visualization of penultimate-layer feature distributions under natural inputs, final PGD-20 adversarial inputs, and intermediate PGD trajectory states. The visualization provides a qualitative comparison between FAT and SAT-IT under adversarial trajectories.

4.4. Ablation

Table 7: Ablation study of SAT-IT components on CIFAR-10 (ResNet-18).

Method	Clean	FGSM	PGD-20	PGD-100	AutoAttack
SAT-IT w/o MI	0.8212	0.5932	0.5621	0.5405	0.4871
SAT-IT w/o SD & $\mathcal{R}_{\text{ent}}$	0.8157	0.5982	0.5642	0.5440	0.4839
SAT-IT w/o SD	0.8181	0.5951	0.5636	0.5452	0.4791
SAT-IT w/o $\mathcal{R}_{\text{ent}}$	0.8158	0.5977	0.5655	0.5468	0.4870
SAT-IT (full)	0.8291	0.6061	0.5727	0.5527	0.4874

As shown in Table 7, we ablate the key components of SAT-IT, including the MI-based weighting, self-distillation (SD), and the entropy regularizer $\mathcal{R}_{\text{ent}}$.

We first examine the core design of SAT-IT, namely the trajectory MI indicator and the resulting stability-aware weighting. Compared with removing MI, the full SAT-IT achieves consistently stronger robustness under iterative PGD attacks (PGD-20/100 improve from 0.5621/0.5405 to 0.5727/0.5527), while maintaining a comparable AutoAttack performance (0.4871 $\rightarrow$ 0.4874). These results indicate that MI-guided weighting effectively emphasizes trajectory-unstable samples and strengthens robustness against strong gradient-based attacks.

We further evaluate auxiliary components, including self-distillation (implemented with EMA-smoothed historical targets) and the entropy regular-

izer $\mathcal{R}_{\text{ent}}$. With these components, SAT-IT improves the overall robustness–accuracy trade-off: compared with the configuration without SD and $\mathcal{R}_{\text{ent}}$, the full model yields a noticeable gain in clean accuracy ($0.8157 \rightarrow 0.8291$) together with improved robustness under PGD-20 ($0.5642 \rightarrow 0.5727$) and a slight improvement under AutoAttack ($0.4839 \rightarrow 0.4874$). Overall, SAT-IT achieves stronger resistance to iterative PGD attacks on top of a modest AA gain, while also delivering a substantial improvement in natural accuracy.

To further verify whether these component-wise effects transfer to a more fine-grained dataset, we include an additional CIFAR-100 ablation study in Table 8. Compared with the variant without MI, the full SAT-IT improves clean accuracy ($0.5605 \rightarrow 0.5650$) and obtains higher FGSM/PGD-20/PGD-100 robustness ($0.3481/0.3337/0.3244 \rightarrow 0.3537/0.3381/0.3288$), showing that trajectory-aware stability weighting remains beneficial on CIFAR-100. Removing self-distillation weakens both clean accuracy and robustness, while removing the entropy regularizer slightly improves AutoAttack in this single run but reduces clean accuracy and PGD robustness. Overall, the full model provides the best clean–PGD trade-off on CIFAR-100.

Table 8: Additional ablation study of SAT-IT components on CIFAR-100 (ResNet-18).

Method	Clean	FGSM	PGD-20	PGD-100	CW _∞	AutoAttack
SAT-IT w/o MI	0.5605	0.3481	0.3337	0.3244	0.2724	0.2551
SAT-IT w/o SD & $\mathcal{R}_{\text{ent}}$	0.5242	0.3400	0.3290	0.3220	0.2772	0.2593
SAT-IT w/o SD	0.5587	0.3459	0.3342	0.3234	0.2704	0.2539
SAT-IT w/o $\mathcal{R}_{\text{ent}}$	0.5385	0.3407	0.3290	0.3229	0.2757	0.2625
SAT-IT (full)	0.5650	0.3537	0.3381	0.3288	0.2797	0.2608

5. Conclusion and Future Work

We propose SAT-IT, a stability-aware adversarial training framework that leverages intermediate trajectories generated during adversarial example construction. Instead of relying only on the final adversarial example, SAT-IT models the prediction dynamics along the PGD trajectory. Specifically, we introduce a MI-based trajectory instability metric to quantify predictive distribution fluctuations across intermediate states, and combine it with a misclassification-event gate to construct adaptive sample-wise weights. In this way, SAT-IT imposes stronger robust constraints on trajectory-unstable

and attack-prone samples, while avoiding excessive regularization on stable samples.

To further stabilize training, SAT-IT incorporates a lightweight self-distillation mechanism based on EMA-smoothed historical targets and stage-wise entropy regularization. The self-distillation component constructs stable soft targets from the model’s own historical predictions, avoiding the need for external teacher models and reducing potential teacher-student mismatch. The entropy regularizer further mitigates over-sharpening caused by stability-aware weighting, helping balance robustness and natural accuracy.

Extensive experiments on CIFAR-10, CIFAR-100, and TinyImageNet-200 demonstrate the effectiveness of SAT-IT under multiple white-box attacks, including FGSM, PGD, CW_∞ , and AutoAttack. Additional black-box transfer evaluations further show that the robustness gains are not limited to the white-box setting. Compared with FAT, which mainly uses intermediate states to select early-stopped friendly adversarial examples, SAT-IT explicitly models the full adversarial trajectory. The post-hoc trajectory MI analysis also shows that SAT-IT produces more stable prediction dynamics along PGD trajectories, supporting the effectiveness of trajectory-level stability modeling.

Currently, SAT-IT is mainly validated under ℓ_∞ -constrained image classification tasks and CNN architectures. Future work could extend SAT-IT to other threat models such as ℓ_2 and spatial perturbations, evaluate its effectiveness on emerging architectures such as Vision Transformers, and explore adaptive trajectory lengths for more efficient stability estimation. It would also be valuable to further investigate the theoretical relationship between trajectory stability, decision-boundary smoothness, and adversarial robustness, as well as the applicability of SAT-IT to larger-scale and multimodal learning scenarios.

CRediT authorship contribution statement

Ziyu Xu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work

reported in this paper.

Data and code availability

The datasets used in this study are publicly available benchmark datasets. The source code for SAT-IT has been released as an open-source implementation at <https://github.com/11260371/SAT-IT>.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used generative AI tools for language polishing, LaTeX formatting assistance, and editorial refinement. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A. Additional Ablation Results

Unless otherwise specified, the supplementary ablation experiments follow the same CIFAR-10 training and evaluation protocol as the main experiments. Robustness is evaluated under FGSM, PGD-20, PGD-100, and AutoAttack when the corresponding records are available; missing entries indicate that the attack evaluation was not conducted for that exploratory run.

Appendix A.1. Sensitivity analysis of EMA coefficients

Table A.9 reports an exploratory sensitivity analysis of the EMA-based self-distillation coefficients. These runs keep the original MART objective structure and do not apply SAT-IT stability-aware weighting to the robust loss. The comparison is used to guide the choice of EMA coefficients in the final SAT-IT configuration.

The results suggest that a larger historical smoothing coefficient is generally more favorable for robustness. For example, $A = 0.9$ achieves stronger

Table A.9: Sensitivity analysis of EMA/self-distillation coefficients on CIFAR-10 with ResNet-18. All runs use $\beta = 5$ and keep the original MART objective structure without SAT-IT stability-aware weighting. “Flip weight” denotes replacing the corresponding EMA coefficient with the flip-instability weight. Missing values indicate that the attack evaluation was not conducted for that exploratory run.

Coefficient	Variant	Clean	FGSM	PGD-20	PGD-100	AutoAttack	Epoch
A	MART+EMA, $A = 0.9$	0.8253	0.5918	0.5552	0.5347	0.4868	81
A	MART+EMA, $A = 0.5$	0.8261	–	0.5530	–	–	81
A	MART+EMA, $A \leftarrow$ flip weight	0.8202	0.5840	0.5524	0.5321	0.4819	76
B	MART+EMA, $B = 0.9$	0.8201	0.5959	0.5635	0.5455	0.4869	90
B	MART+EMA, $B = 0.5$	0.8253	0.5918	0.5552	0.5347	0.4868	81
B	MART+EMA, $B = 0.1$	0.8402	–	0.5410	–	–	79
B	MART+EMA, $B \leftarrow$ flip weight	0.8396	0.5957	0.5524	0.5265	0.4736	81
C	MART+EMA, $C = 0.5$	0.8253	0.5918	0.5552	0.5347	0.4868	81
C	MART+EMA, $C = 0.1$	0.8361	–	0.5575	–	–	90
C	MART+EMA, $C \leftarrow$ flip weight	0.8283	0.5907	0.5553	0.5319	0.4846	81

PGD-20/PGD-100 and AutoAttack performance than replacing A with the flip-instability weight, while $B = 0.9$ gives the best PGD-20 and PGD-100 robustness in the B group. In contrast, directly replacing EMA coefficients with flip-instability weights tends to improve or preserve clean accuracy in some cases but weakens robustness under stronger attacks. This supports using stable EMA-based soft targets rather than directly injecting flip-based weights into the self-distillation coefficients.

Appendix A.2. Effect of regularization strength

Table A.10 evaluates the effect of different regularization strengths in SAT-IT. All runs are based on MART+EMA with $(A, B, C) = (0.9, 0.9, 0.1)$ and SAT-IT stability-aware weighting. We vary the regularization strength λ after epoch 75 and compare natural-prediction-based and adversarial-prediction-based regularization targets.

The results show that SAT-IT is not highly sensitive to the post-75-epoch value of λ within a moderate range. Using adversarial predictions with $\lambda = 0.6$ gives the best PGD-100 robustness in this group, while using natural predictions with $\lambda = 0.6$ achieves the highest clean accuracy and competitive PGD-20/AA performance. We therefore use these results as supplementary evidence that the regularization target and strength mainly affect the clean-robustness balance rather than changing the overall robustness trend.

Table A.10: Effect of different regularization strengths λ on CIFAR-10 with ResNet-18. All runs use $\beta = 5$ and MART+EMA with $(A, B, C) = (0.9, 0.9, 0.1)$ combined with SAT-IT stability-aware weighting.

Later target	λ (epoch ≤ 75)	λ (epoch > 75)	Clean	FGSM	PGD-20	PGD-100	AA	Epoch
Nat. probs	1.0	1.0	0.8258	0.6024	0.5652	0.5453	0.4885	91
Adv. probs	1.0	0.7	0.8259	0.6038	0.5672	0.5473	0.4868	91
Adv. probs	1.0	0.6	0.8227	0.6015	0.5672	0.5487	0.4872	93
Adv. probs	1.0	0.5	0.8234	0.5989	0.5674	0.5475	0.4843	90
Adv. probs	1.0	0.3	0.8211	0.6013	0.5664	0.5449	0.4875	94
Nat. probs	1.0	0.6	0.8281	0.6021	0.5670	0.5454	0.4873	94

Appendix A.3. Comparison of instability-aware regularization objectives

Table A.11 compares several instability-aware regularization objectives under the SAT-IT weighting scheme. These experiments are used to analyze the effect of the auxiliary regularization objective while keeping the same MART+EMA backbone and stability-aware weighting design.

Table A.11: Effect of different instability-aware regularization objectives on CIFAR-10 with ResNet-18. All runs use SAT-IT stability-aware weighting.

Regularization objective	Clean	PGD-20	Epoch
Instability-aware L2 consistency regularization	0.8103	0.5652	92
Instability-aware KL consistency regularization	0.8101	0.5643	90
Instability-aware Jensen–Shannon consistency regularization	0.8091	0.5634	90
Instability-aware gradient norm penalty	0.8145	0.5638	90
Instability-aware entropy regularization (natural predictions)	0.8258	0.5652	91
Instability-aware negative entropy regularization (adversarial predictions)	0.7778	0.5493	77
L2 parameter drift regularization	0.8018	0.5656	90

Among the compared objectives, natural-prediction entropy regularization provides the best clean accuracy while maintaining competitive PGD-20 robustness. Although L2 parameter drift regularization obtains a slightly higher PGD-20 score, its clean accuracy is much lower. This supports using entropy regularization based on natural predictions as a balanced auxiliary objective for SAT-IT.

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